

Exercise 17

Chapter 2, Page 76

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Introduction to Electrodynamics

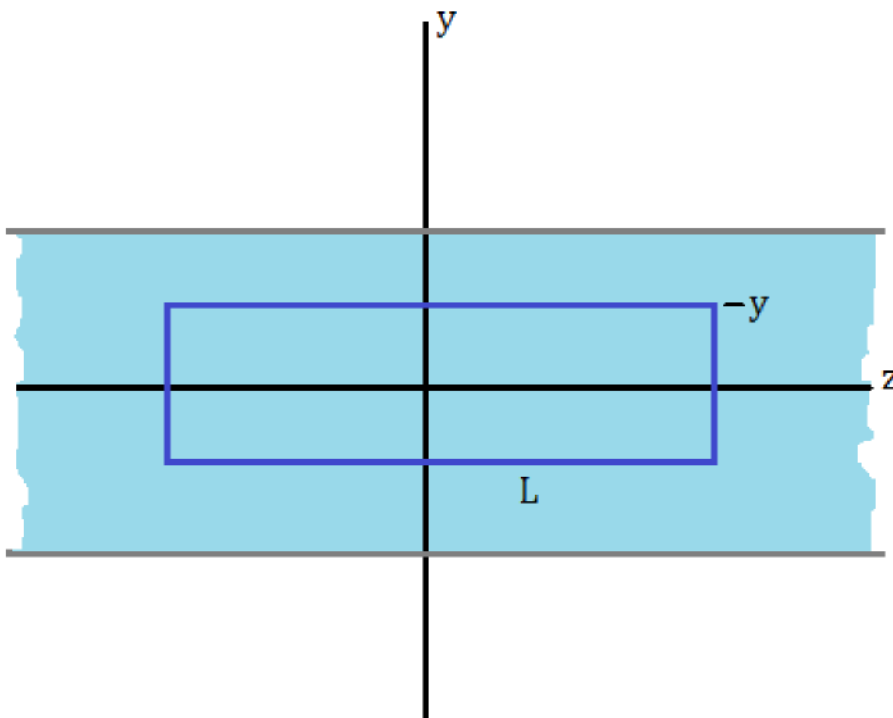
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Step 1

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Choose for the Gaussian surface the pillbox of thickness $2y$. By symmetry, the electric field will only contribute flux to the upper and lower side of the pillbox.



Step 2

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If $|y| < d$ the charge enclosed is:

$$Q_{enc} = \rho L^2(2y)$$

The electric field is perpendicular to the xz plane and is equal to:

$$2EL^2 = \frac{\rho L^2(2y)}{\epsilon_0} \implies \vec{E} = \boxed{\pm \frac{\rho y}{\epsilon_0} \hat{y}}$$

If $|y| > d$ then the charge enclosed, and the electric field are:

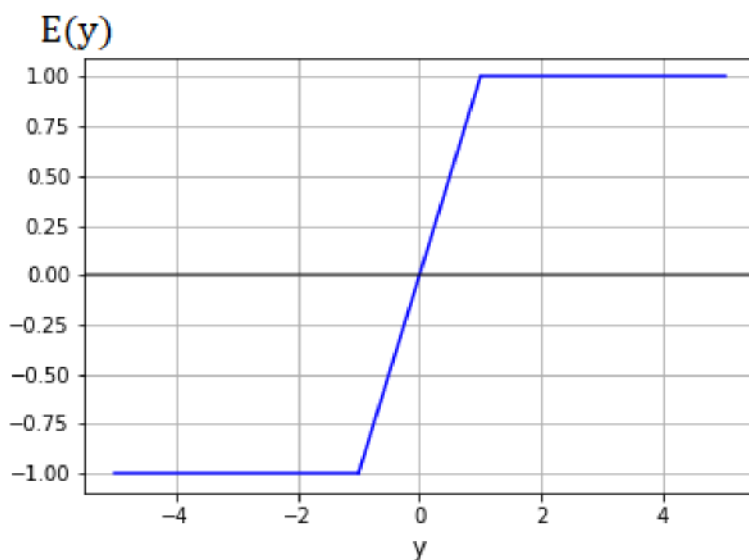
$$Q_{enc} = L^2 \rho(2d)$$

$$2EL^2 = \frac{L^2 \rho(2d)}{\epsilon_0} \implies \vec{E} = \boxed{\pm \frac{\rho d}{\epsilon_0} \hat{y}}$$

, and is constant in magnitude, just like for the case of the infinite plane of charge. The diagram is given below, for $\rho d / \epsilon_0 = 1$.

Step 3

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$$\vec{E} = \begin{cases} \pm \frac{\rho y}{\epsilon_0} \hat{y} & , |y| < d \\ \pm \frac{\rho d}{\epsilon_0} \hat{y} & , |y| > d \end{cases}$$

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